Swagger Supported by SMARTBEAR

http://127.0.0.1:8000/swagger/?format=oper

Explore

TfL Analytics API v1 OAS 2.0

[Base URL: 127.0.0.1:8000/api]
http://127.0.0.1:8000/swagger/?format=openapi

API for user routes, favourite stopss, incidents, and TfL analytics. Completed as a coursework requirement for COMP3011 Web Services and Web Data.

Schemes

HTTP

Django Login

Authorize

Filter by tag

Analytics - Lines

GET /analytics/lines/{line_id}/headway/ analytics_lines_headway_list

Compute average headway for a line

Computes the average headway (time between consecutive services) for a line.

Parameters

Try it out

Name	Description
line_id * required string (path)	line_id

Responses

Response content type application/json

Code	Description
200	Successfully computed average headway for the line

Example Value

```
{
  "line_id": "victoria",
  "average_headway_seconds": 180
}
```

404	Line not found or has insufficient arrival data

Example Value

```
{
  "detail": "Line 'southern' not found or has insufficient arrival data."
}
```

GET `/analytics/lines/{line_id}/incident-status/`

Line status based on user incidents

`analytics_lines_incident-status_list`

Computes the operational status of a line using recent user-reported incidents. The status is categorized based on the severity and number of incidents reported in the last 30 minutes.

Parameters

Try it out

Name	Description
line_id * required `string` *(path)*	line_id

Responses

Response content type: application/json

Code	Description
200	Incident-based line status

Example Value

```
{
  "line_id": "victoria",
  "status": "Moderate delays",
  "score": 5
}
```

400 Error: Bad Request - Invalid Line ID

Example Value

```
{
  "error": "Invalid Line ID"
}
```

Analytics - Stops

GET `/analytics/stops/{stop_id}/average-wait/` `analytics_stops_average-wait_list`

Compute average wait time for a stop

Computes the average wait time (in seconds) for a specific stop based on current arrival data for the station (i.e., irrespective of the line, calculates the average wait).

Parameters Try it out

Name	Description
stop_id * required `string` *(path)*	stop_id

Responses Response content type application/json

Code	Description
200	Average wait time for the stop

Example Value

```
{
  "stop_id": "940GZZLUBST",
  "average_wait_seconds": 142.5
```

```
}
```

400 Invalid stop ID

Example Value

```
{
  "error": "Invalid stop ID"
}
```

404 Stop has no arrival data

Example Value

```
{
  "detail": "Stop 'HUBZMG' not found or has no arrival data."
}
```

GET `/analytics/stops/{stop_id}/incident-status/`

Stop status based on user incidents

`analytics_stops_incident-status_list`

Computes the operational status of a stop using recent user-reported incidents. The status is categorized based on the severity and number of incidents reported in the last 30 minutes.

Parameters Try it out

Name	Description
stop_id * required `string` *(path)*	stop_id

Responses Response content type application/json

Code	Description
200	Incident-based stop status

Example Value

```
{
  "stop_id": "940GZZLUBST",
```

```
  "status": "Moderate issues",
  "score": 5
}
```

400 Error: Bad Request - Invalid Stop ID

Example Value

```
{
  "error": "Invalid Stop ID"
}
```

User Incidents

GET `/incidents/` Retrieve all user-reported incidents `incidents_list`

Retrieves all user-reported incidents.

Parameters Try it out

No parameters

Responses Response content type application/json

Code	Description
200	List of user incidents. Returns empty list if no issues reported.

Example Value

```
[
  {
    "id": 1,
    "user": "John Doe",
    "stop": "Whitechapel (HUBZWL)",
    "line": "Elizabeth line",
    "description": "Severe delays, platform overcrowded",
    "severity": 4,
    "created_at": "2026-03-12T17:30:00Z"
  },
  {
    "id": 2,
    "user": "Jane Smith",
    "stop": "Liverpool Street (HUBZLST)",
```

```
    "line": "Central line",
    "description": "Minor delays, train at full capacity",
    "severity": 2,
    "created_at": "2026-03-12T16:45:00Z"
  }
]
```


POST `/incidents/` Report a user incident `incidents_create`

Creates a new incident report for a stop and line.

Parameters Try it out

Name	Description
data * required object *(body)*	Example Value | **Model**

UserIncident {

- **stop*** — **Stop** string
  title: Stop
  maxLength: 20
  minLength: 1
  Stop ID where the incident occurred
- **line*** — **Line** string
  title: Line
  maxLength: 20
  minLength: 1
  Line ID affected by the incident
- **description*** — **Description** string
  title: Description
  maxLength: 200
  minLength: 1
  Description of the incident
- **severity*** — **Severity** integer
  title: Severity
  maximum: 5
  minimum: 1
  Severity level (1-5)

}

Responses Response content type application/json

Code	Description
201	Incident created

Example Value

```
{
  "id": 3,
  "user": "John Doe",
  "stop": "Whitechapel (HUBZWL)",
  "line": "Elizabeth line",
  "description": "Severe delays, platform overcrowded",
  "severity": 4
}
```

400	Invalid input data

Example Value

```
{
  "line": "Line ID does not exist."
}
```

GET `/incidents/{id}/` Retrieve a specific incident `incidents_read`

Retrieve a specific incident from a given Incident ID. Example: ID = 1

Parameters Try it out

Name	Description
id * required `integer` *(path)*	A unique integer value identifying this user incident. id

Responses Response content type application/json

Code	Description

200 Incident details

Example Value

```
{
  "id": 1,
  "user": "John Doe",
  "stop": "Whitechapel (HUBZWL)",
  "line": "Elizabeth line",
  "description": "Severe delays, platform overcrowded",
  "severity": 4,
  "created_at": "2026-03-12T17:30:00Z"
}
```

404 Incident not found

Example Value

```
{
  "detail": "No UserIncident matches the given query."
}
```

PUT `/incidents/{id}/` Update a specific incident `incidents_update`

Update a specific incident using a given Incident ID

Parameters Try it out

Name	Description
data * required `object` *(body)*	Example Value | **Model**

UserIncident ⌄ {

- **stop*** — **Stop** ⌄ `string`
  title: Stop
  maxLength: 20
  minLength: 1
  Stop ID where the incident occurred
- **line*** — **Line** ⌄ `string`
  title: Line
  maxLength: 20
  minLength: 1
  Line ID affected by the incident
- **description*** — **Description** ⌄ `string`
  title: Description
  maxLength: 200
  minLength: 1
  Description of the incident
- **severity*** — **Severity** ⌄ `integer`
  title: Severity
  maximum: 5
  minimum: 1
  Severity level (1-5)

}

id * required
`integer`
(path)

A unique integer value identifying this user incident.

id

Responses

Response content type: application/json ⌄

Code	Description
200	Incident details after update.

Example Value

```
{
  "id": 1,
  "user": "John Doe",
  "stop": "Whitechapel (HUBZWL)",
  "line": "Elizabeth line",
  "description": "Severe delays, platform overcrowded",
  "severity": 4,
  "created_at": "2026-03-12T17:30:00Z"
}
```

400 Error: Bad Request (Invalid input data - full request body required)

Example Value

```
{
  "stop": [
    "Stop ID is required."
  ]
}
```

404 Incident not found

Example Value

```
{
  "detail": "No UserIncident matches the given query."
}
```

PATCH `/incidents/{id}/` `incidents_partial_update`

Partially update a specific incident

Partially update a specific incident using a given Incident ID. Only the updated fields need to be provided in the request body.

Parameters Try it out

Name	Description
data * required `object` *(body)*	Example Value | **Model**

UserIncident ⌄ {

stop*	**Stop** ⌄ string *title: Stop* *maxLength: 20* *minLength: 1* Stop ID where the incident occurred
line*	**Line** ⌄ string *title: Line* *maxLength: 20* *minLength: 1* Line ID affected by the incident
description*	**Description** ⌄ string *title: Description* *maxLength: 200* *minLength: 1* Description of the incident
severity*	**Severity** ⌄ integer *title: Severity* *maximum: 5* *minimum: 1* Severity level (1-5)

}

id * required

integer

(path)

A unique integer value identifying this user incident.

id

Responses

Response content type **application/json** ⌄

Code	Description
200	Incident details after partial update. **Example Value** |

```
{
  "id": 1,
  "user": "John Doe",
  "stop": "Whitechapel (HUBZWL)",
  "line": "Elizabeth line",
  "description": "Severe delays, platform overcrowded",
  "severity": 4,
  "created_at": "2026-03-12T17:30:00Z"
}
```

404	Incident not found

Example Value

```
{
  "detail": "No UserIncident matches the given query."
}
```

DELETE `/incidents/{id}/` Delete a specific incident `incidents_delete`

Delete a specific incident via a given Incident ID

Parameters Try it out

Name	Description
id * required `integer` *(path)*	A unique integer value identifying this user incident. id

Responses Response content type application/json

Code	Description
204	Incident deleted successfully
404	Incident not found

Example Value

```
{
  "detail": "No UserIncident matches the given query."
}
```

Lines

GET `/lines/` Retrieve all lines in the database `lines_list`

Retrieve all lines in the database.

Parameters Try it out

No parameters

Responses Response content type application/json

Code	Description
200	List of lines

Example Value


```
[
  {
    "line_id": "bakerloo",
    "name": "Bakerloo",
    "mode": "tube"
  },
  {
    "line_id": "central",
    "name": "Central",
    "mode": "tube"
  },
  {
    "line_id": "circle",
    "name": "Circle",
    "mode": "tube"
  }
]
```


User Routes

GET `/routes/` Retrieve all user routes `routes_list`

Retrieve all user-added/favourite routes.

Parameters Try it out

No parameters

Responses Response content type application/json

Code	Description
200	List of user routes

Example Value

```
[
  {
    "id": 1,
    "user": "John Doe",
    "from_stop": "Farringdon (HUBZFD)",
    "to_stop": "Whitechapel (HUBZWL)",
    "line": "Elizabeth line"
  },
  {
    "id": 2,
    "user": "Jane Smith",
    "from_stop": "Ealing Broadway (HUBEAL)",
    "to_stop": "Liverpool Street (HUBZLST)",
    "line": "Central line"
  }
]
```

POST `/routes/` Add a new favourite route `routes_create`

Adds a new route for a user, including origin stop, destination stop, and line.

Parameters Try it out

Name	Description
data * required `object` *(body)*	Example Value | **Model**

```
UserRoute ⌄ {
    from_stop*        From stop ⌄ string
                      title: From stop
                      maxLength: 20
                      minLength: 1

                      Origin stop ID

    to_stop*          To stop ⌄ string
                      title: To stop
                      maxLength: 20
                      minLength: 1

                      Destination stop ID

    line*             Line ⌄ string
                      title: Line
                      maxLength: 20
                      minLength: 1

                      Line ID for the route

}
```

Responses Response content type application/json

Code	Description
201	Route created successfully

Example Value

```
{
  "id": 12,
  "user": "John Doe",
  "from_stop": "Farringdon (HUBZFD)",
  "to_stop": "Whitechapel (HUBZWL)",
  "line": "Elizabeth line"
}
```

400	Invalid input data

Example Value

```
{
  "from_stop": "Stop ID does not exist."
}
```

GET /routes/{id}/ Retrieve a user route routes_read

Retrieve a specific user route by its ID.

Parameters Try it out

Name	Description
id * required `integer` *(path)*	A unique integer value identifying this user route. id

Responses Response content type application/json

Code	Description
200	User route details **Example Value** `{ "id": 1, "user": "John Doe", "from_stop": "Farringdon (HUBZFD)", "to_stop": "Whitechapel (HUBZWL)", "line": "Elizabeth line" }`
404	Route not found **Example Value** `{ "error": "User route not found" }`

PUT `/routes/{id}/` Update a user route `routes_update`

Update a specific user route

Parameters Try it out

Name	Description
data * required object *(body)*	Example Value | **Model** **UserRoute** {

```
UserRoute ⌄ {
    from_stop*        From stop ⌄ string
                      title: From stop
                      maxLength: 20
                      minLength: 1

                      Origin stop ID

    to_stop*          To stop ⌄ string
                      title: To stop
                      maxLength: 20
                      minLength: 1

                      Destination stop ID

    line*             Line ⌄ string
                      title: Line
                      maxLength: 20
                      minLength: 1

                      Line ID for the route

}
```

id * required integer *(path)*	A unique integer value identifying this user route. id

Responses Response content type **application/json**

Code	Description
200	Route updated successfully **Example Value** |

```
{
  "id": 1,
  "user": "John Doe",
  "from_stop": "Farringdon (HUBZFD)",
  "to_stop": "Whitechapel (HUBZWL)",
  "line": "Elizabeth line"
}
```

400	Invalid input data

Example Value

```
{
  "from_stop": "Stop ID does not exist."
}
```

404 Route not found

Example Value

```
{
  "detail": "No UserRoute matches the given query."
}
```

PATCH `/routes/{id}/` Partially update a user route `routes_partial_update`

Partially update a specific user route. Only the updated fields need to be provided in the request body.

Parameters Try it out

Name	Description
data * required `object` *(body)*	Example Value | **Model**

UserRoute {

- `from_stop*` — **From stop** `string`
  title: From stop
  maxLength: 20
  minLength: 1
  `Origin stop ID`
- `to_stop*` — **To stop** `string`
  title: To stop
  maxLength: 20
  minLength: 1
  `Destination stop ID`
- `line*` — **Line** `string`
  title: Line
  maxLength: 20
  minLength: 1
  `Line ID for the route`

}

id * required
integer
(path)

A unique integer value identifying this user route.

id

Responses Response content type application/json

Code	Description
200	Route partially updated successfully

Example Value

```
{
  "id": 1,
  "user": "John Doe",
  "from_stop": "Farringdon (HUBZFD)",
  "to_stop": "Whitechapel (HUBZWL)",
  "line": "Elizabeth line"
}
```

400	Invalid input data

Example Value

```
{
  "from_stop": "Stop ID does not exist."
}
```

404	Route not found

Example Value

```
{
  "detail": "No UserRoute matches the given query."
}
```

DELETE /routes/{id}/ Delete a user route routes_delete

Delete a specific user route

Parameters Try it out

Name	Description
id * required `integer` *(path)*	A unique integer value identifying this user route. id

Responses Response content type application/json

Code	Description
204	Route deleted successfully
404	Route not found **Example Value** `{ "detail": "No UserRoute matches the given query." }`

User Stops

GET `/stations/` Retrieve all favourite stops `stations_list`

Retrieve all favourite stops for all users

Parameters Try it out

No parameters

Responses Response content type application/json

Code	Description
200	List of user favourite stops. Returns empty list if no favourite stops added.

Example Value

```
[
  {
    "id": 1,
    "user": "John Doe",
    "stop": "Farringdon (HUBZFD)"
  },
  {
    "id": 2,
    "user": "Jane Smith",
    "stop": "Whitechapel (HUBZWL)"
  }
]
```

POST `/stations/` Add a favourite stop `stations_create`

Marks a stop as a favourite for a user.

Parameters Try it out

Name	Description
data * required object *(body)*	Example Value | **Model** **UserStation** { stop* **Stop** string *title: Stop* *maxLength: 20* *minLength: 1* Stop ID to mark as favourite }

Responses Response content type application/json

Code	Description
201	Favourite stop added Example Value

```
{
  "id": 5,
  "user": "John Doe",
  "stop": "Farringdon (HUBZFD)"
}
```

GET `/stations/{id}/` Retrieve a specific favourite stop `stations_read`

Retrieve a specific favourite stop for a user

Parameters Try it out

Name	Description
id * required `integer` *(path)*	A unique integer value identifying this user station. id

Responses Response content type application/json

Code	Description
200	Favourite stop details **Example Value** `{ "id": 1, "user": "John Doe", "stop": "Farringdon (HUBZFD)" }`
404	Favourite stop not found **Example Value** `{ "detail": "No UserStation matches the given query." }`

DELETE `/stations/{id}/` Delete a favourite stop `stations_delete`

Delete a specific favourite stop for a user

Parameters Try it out

Name	Description
id * required `integer` *(path)*	A unique integer value identifying this user station. id

Responses Response content type application/json

Code	Description
204	Favourite stop deleted successfully
404	Favourite stop not found **Example Value** `{ "detail": "No UserStation matches the given query." }`

Stops

GET `/stops/` Retrieve all stops in the database `stops_list`

Retrieve all stops in the database. Supports filtering by mode (e.g. tube, elizabeth-line, dlr, overground, tram).

Parameters Try it out

Name	Description
mode `string` *(query)*	mode

Responses Response content type application/json

Code	Description
200	List of stops as per filter. (Example filter: "elizabeth-line")

Example Value


```
[
  {
    "stop_id": "HUBBDS",
    "name": "Bond Street",
    "mode": "elizabeth-line",
    "lat": 51.513362,
    "lon": -0.148795
  },
  {
    "stop_id": "HUBCAW",
    "name": "Canary Wharf",
    "mode": "elizabeth-line",
    "lat": 51.503734,
    "lon": -0.019121
  },
  {
    "stop_id": "HUBEAL",
    "name": "Ealing Broadway",
    "mode": "elizabeth-line",
    "lat": 51.514993,
    "lon": -0.302131
  }
]
```


GET `/stops/{stop_id}/` Retrieve a stop by ID `stops_read`

Retrieve detailed information about a specific stop using its stop ID. Returns name, mode, and coordinates. Useful for validating stop IDs or displaying stop metadata.

Parameters Try it out

Name	Description
stop_id * required string (path)	stop_id

Responses Response content type application/json

Code	Description
200	Stop found

Example Value

```
{
  "stop_id": "HUBBDS",
  "name": "Bond Street",
  "mode": "elizabeth-line",
  "lat": 51.513362,
  "lon": -0.148795
}
```

404	Stop not found

Example Value

```
{
  "error": "Stop ID not found"
}
```

Arrivals

GET /stops/{stop_id}/arrivals/ stops_arrivals_list

Retrieve arrival predictions for a stop

Retrieves all previously saved arrival predictions for a stop. Use refresh = true to fetch and additionally return fresh data from TfL.

Parameters Try it out

Name	Description

refresh
boolean
(query)

If true, fetch fresh arrival data from TfL before returning results.

--

stop_id * required
string
(path)

stop_id

Responses

Response content type: application/json

Code	Description
200	List of arrival predictions

Example Value


```
[
  {
    "id": 2627,
    "direction": "outbound",
    "destination_name": "Hainault Underground Station",
    "predicted_time": "2026-03-12T18:11:37Z",
    "time_to_station": 1611,
    "recorded_at": "2026-03-12T17:44:54.319715Z",
    "stop": "940GZZLUNBP",
    "line": "central"
  },
  {
    "id": 2628,
    "direction": "outbound",
    "destination_name": "Hainault Underground Station",
    "predicted_time": "2026-03-12T17:45:38Z",
    "time_to_station": 52,
    "recorded_at": "2026-03-12T17:44:54.323132Z",
    "stop": "940GZZLUNBP",
    "line": "central"
  },
  {
    "...": "..."
  }
]
```


400	Error: Bad Request - Invalid Stop ID

Example Value


```
{
  "error": "Invalid Stop ID"
}
```

Models

UserIncident {

- **id** — ID integer
  title: ID
  readOnly: true
- **user** — User string
  title: User
  readOnly: true
  minLength: 1
- **stop*** — Stop string
  title: Stop
  maxLength: 20
  minLength: 1

  Stop ID where the incident occurred
- **line*** — Line string
  title: Line
  maxLength: 20
  minLength: 1

  Line ID affected by the incident
- **description*** — Description string
  title: Description
  maxLength: 200
  minLength: 1

  Description of the incident
- **severity*** — Severity integer
  title: Severity
  maximum: 5
  minimum: 1

  Severity level (1-5)
- **created_at** — Created at string($date-time)
  title: Created at
  readOnly: true

}

UserRoute ⌄ {

id	**ID** ⌄ integer *title: ID* *readOnly: true*
user	**User** ⌄ string *title: User* *readOnly: true* *minLength: 1*
from_stop*	**From stop** ⌄ string *title: From stop* *maxLength: 20* *minLength: 1* Origin stop ID
to_stop*	**To stop** ⌄ string *title: To stop* *maxLength: 20* *minLength: 1* Destination stop ID
line*	**Line** ⌄ string *title: Line* *maxLength: 20* *minLength: 1* Line ID for the route

}

UserStation ⌄ {

id	**ID** ⌄ integer *title: ID* *readOnly: true*
user	**User** ⌄ string *title: User* *readOnly: true* *minLength: 1*
stop*	**Stop** ⌄ string *title: Stop* *maxLength: 20* *minLength: 1* Stop ID to mark as favourite

}